Deep Learning Systems Analysis Report

Deep Learning for Player Action Prediction from Atari Gameplay Frames

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Report Overview

This report describes a deep learning experiment that trains convolutional neural networks to predict player actions from Atari Montezuma's Revenge gameplay frames. The central research question is whether providing a model with four consecutive frames of temporal context improves action prediction over a single static frame. Two models are compared under a controlled experimental design in which exactly one variable changes: the input representation. A third experiment using a boosted class weight is also described and analyzed. The report covers the dataset, model architecture and design decisions, training methodology, evaluation results, limitations, and responsible use considerations.

Dataset and Task Description

The dataset is Atari-HEAD (Zhang et al., 2020), a collection of real human gameplay recordings from Atari 2600 games. Atari-HEAD captures synchronized eye-tracking data, gameplay frames, and action inputs from a single human player across multiple play sessions. The Montezuma's Revenge subset used in this project contains 27 sessions (20 standard and 7 highscore) totaling 447,300 frames. Each frame is paired with the action the player pressed at that moment, recorded as an Arcade Learning Environment (ALE) action code. The dataset is publicly available under CC BY 4.0 and was retrieved from Zenodo.

Montezuma's Revenge was selected over other Atari titles for three reasons. First, it is a side scrolling platformer requiring jumping, ladder climbing, key collection, and hazard avoidance, making its action vocabulary directly relevant to platformer behavior modeling. Second, its class distribution is far more balanced than alternatives such as Breakout, where the NOOP action accounts for over 78% of frames. Third, its 447,300 frame count provides sufficient scale for deep learning training on GPU hardware.

ALE action codes were mapped to five simplified classes representing the meaningful player actions in the game: NOOP (idle or fire with no direction), RIGHT, LEFT, UP (climb up or jump), and DOWN (crouch or climb down). The resulting class distribution is: LEFT 27.8%, NOOP 26.4%, RIGHT 25.7%, DOWN 10.6%, UP 9.5%. The dataset represents a single player's behavioral signature across many play sessions at varying skill levels.

All 447,300 frames are stored as 84x84 grayscale uint8 arrays in CPU RAM (approximately 2.94 GB), with normalization to float32 deferred to batch load time to avoid an out of memory condition during dataset construction.

Model Architecture and Design Decisions

Architecture

Both the baseline and experimental models use the same ActionCNN architecture, derived from the DQN convolutional design (Mnih et al., 2015). The architecture consists of three convolutional layers followed by two fully connected layers:

- Conv2d(in_channels, 32, kernel 8x8, stride 4) + ReLU
- Conv2d(32, 64, kernel 4x4, stride 2) + ReLU
- Conv2d(64, 64, kernel 3x3, stride 1) + ReLU
- Linear(3136, 512) + ReLU
- Dropout(p=0.5)
- Linear(512, 5)

At 84x84 input, the three convolutional layers reduce spatial dimensions to a 7x7x64 feature map, which flattens to 3,136 features before the classification head. The flat feature size is computed dynamically, making the class portable to other input resolutions. The baseline model (in_channels=1) has 1,680,549 parameters. The experimental model (in_channels=4) has 1,686,693 parameters.

The DQN architecture was chosen because it was designed specifically for Atari frame input and is the established benchmark in the Atari deep learning literature. Its translation invariant convolutional filters are well suited to detecting local spatial features such as ball position, ladder edges, and character posture at this resolution.

Input Representation

The baseline model accepts a single 84x84 grayscale frame (in_channels=1), providing only a static snapshot of game state. The experimental model accepts four consecutive frames stacked along the channel dimension (in_channels=4), providing the model with implicit information about motion direction and speed. This multi frame input approach was introduced by Mnih et al. (2015) as a way to encode short range temporal dynamics without requiring a recurrent component.

Training Configuration

Both models are trained with the following fixed settings:

- Optimizer: Adam, learning rate 1e-4 (Kingma & Ba, 2015)
- L2 regularization: weight decay 1e-4
- Loss: class weighted CrossEntropyLoss, weights inversely proportional to class frequency (He & Garcia, 2009)
- Regularization: Dropout p=0.5 between fully connected layers (Srivastava et al., 2014)
- Epochs: 30
- Batch size: 64
- Hardware: NVIDIA GeForce RTX 3080 (CUDA 12.4)

Data Splits

Sessions are split at the session level to prevent data leakage between training and evaluation. No frame from any session appears in more than one split. The 19/4/4 session allocation produces

approximately 70/15/15 proportions by frame count: 314,929 training frames, 65,620 validation frames, and 66,751 test frames.

Class weights are computed from training labels only using inverse frequency weighting, ensuring the minority UP and DOWN classes receive proportional gradient influence without any information from validation or test labels.

Regularization

An initial 30 epoch training run revealed clear overfitting: training loss fell to 0.12 while validation loss diverged to 0.50 by epoch 30, and the train/validation accuracy gap exceeded 7 percentage points. Dropout (p=0.5) and L2 weight decay (1e-4) were added to address this, reducing the gap to approximately 4 percentage points and stabilizing validation loss.

Experimental Comparison

The controlled comparison tests one hypothesis: does four frame temporal context improve action prediction over a single static frame?

What changes: The input channel count increases from 1 to 4. Four consecutive gameplay frames are stacked along the channel dimension. The SessionAwareStackedDataset ensures that four frame windows never span session boundaries, preventing a frame from the end of one session from being paired with frames from the start of the next.

What stays constant: Architecture depth, kernel sizes, strides, hidden layer size, dropout rate, optimizer, weight decay, loss function, class weights, number of epochs, batch size, and the train/validation/test split.

Why this comparison is valid: Changing exactly one variable between two otherwise identical models isolates the effect of that variable. Any difference in test set performance can be attributed to temporal context rather than architectural or training differences.

Experiment 2: Boosted NOOP class weight. After evaluating the baseline and stacked models, analysis of the confusion matrix revealed that NOOP had the highest false negative count in both models, nearly double that of any other class. A third model was trained using the baseline architecture with the NOOP class weight multiplied by 2.0 above its balanced value, all other settings held constant. This experiment tests whether a targeted weight adjustment can improve NOOP recall without reducing overall performance.

Results and Interpretation

Model Comparison

Metric	Baseline (1-frame)	NOOP Boosted (1-frame)	Stacked (4-frame)
Accuracy	87.3%	87.2%	91.5%
Macro F1	87.5%	87.5%	91.8%

Per Class F1 (Stacked Model)

Class	Precision	Recall	F1
NOOP	0.933	0.871	0.901
RIGHT	0.904	0.923	0.913
LEFT	0.925	0.925	0.925
UP	0.894	0.953	0.922
DOWN	0.896	0.960	0.927

Interpretation

The four frame stacked model outperforms the single frame baseline by 4.2 percentage points on both accuracy and macro F1. The near identical accuracy and macro F1 scores for both models confirm that the class distribution is balanced enough that neither metric is misleading on its own.

The improvement is consistent across all five action classes. The largest gains are on the vertical actions: UP recall improved from 0.92 to 0.95 and DOWN recall from 0.92 to 0.96. This directly supports the hypothesis that temporal context helps most for motion dependent actions. A single static frame cannot distinguish a player who is currently climbing a ladder from one who is standing next to it; four consecutive frames encode the directional trajectory that resolves the ambiguity. RIGHT showed the largest F1 improvement (+5.3pp), reflecting the same principle: horizontal direction of travel is encoded in motion rather than any single frame.

The NOOP boosted experiment produced exactly the precision-recall tradeoff expected: NOOP recall improved from 0.857 to 0.901 (+4.3pp) while NOOP precision fell from 0.908 to 0.860 (-4.8pp). The net effect on NOOP F1 was negligible (-0.2pp) and overall accuracy and macro F1 were essentially unchanged. Boosting the class weight moved the model along the precision-recall curve but did not escape it. This empirically confirms that flat five class classification creates a structural tension between NOOP and the directional classes that cannot be resolved through weight adjustment alone.

Non Technical Summary

The model learns to watch Atari gameplay and predict what button the player is pressing. When the model sees only a single screenshot, it achieves about 87% accuracy: it knows roughly what the player is doing, but it cannot tell which direction they are moving from a frozen image. When the model sees four screenshots in sequence, accuracy rises to about 92%. The improvement is largest for up and down movements, which require recognizing that the player is actively traversing a ladder or rope rather than simply standing near one. The gap between one frame and four frame input is the gap between reading a photograph and watching a short video clip.

Error Analysis

The stacked model misclassifies 8.5% of test frames (5,649 of 66,739), compared to the baseline's 12.7% error rate. The confusion patterns are highly structured.

NOOP is the single largest source of remaining errors. NOOP predicted as RIGHT and NOOP predicted as LEFT are the top two confusion pairs, together accounting for roughly 32.5% of all

errors. An idle player standing in a neutral pose near a platform edge produces frames that can resemble the early frames of a horizontal movement, and even four consecutive stationary frames can look like the startup phase of directional movement.

RIGHT to LEFT confusion, the dominant error mode in the baseline, is substantially reduced in the stacked model. Four consecutive frames encode direction of travel, resolving most of the ambiguity a static snapshot cannot. Cases that remain are likely mid reversal frames where the player stops one direction and begins another within the four frame window.

UP and DOWN errors are rare and do not appear among the top confusion pairs, consistent with recall scores of 0.95 and 0.96. Ladder traversal involves distinctive postures and positions that are visually separable even at 84x84 resolution.

The remaining errors in the stacked model are concentrated at the NOOP vs action boundary rather than among directional distinctions. A hierarchical cascade architecture (Stage 1: binary idle vs. active; Stage 2: four class directional) would target this specific failure pattern by giving each stage a simpler and better defined decision boundary.

Limitations and Risks

Single subject data. All 447,300 frames represent one player's behavioral signature. The model learns this player's specific tendencies: their reaction timing, preferred routes, and action patterns. Performance on frames from a different player is unknown and likely lower. Generalization across players would require a multi subject dataset.

Static session split. The 19/4/4 session split was generated with a fixed random seed. Different random seeds could produce different splits, and performance may vary across splits. No cross validation was performed due to training time constraints.

Short temporal window. Four consecutive frames encode motion over a brief window at approximately 60fps (roughly 67ms). Longer behavioral sequences such as planning a route across multiple rooms, responding to enemy positions, or recovering from a fall are not captured by the four frame window. An LSTM applied on top of CNN features could address longer range dependencies.

Resolution and information loss. Resizing frames from 160x210 to 84x84 discards fine grained spatial detail. Some visual cues such as enemy facial direction or subtle platform edge geometry may be lost at this resolution.

Temporal label alignment. The Atari-HEAD label file records the action pressed at the time each frame was captured. There may be a small reaction time offset between the player's visual perception and their button input, introducing label noise into the training data.

Ethical and Responsible Use

Behavioral surveillance concern. A model trained to predict player actions from gameplay frames could in principle be used to profile individual players at a level of granularity beyond what players consent to when playing a game. Applying this technique at scale to monitor or rank players without disclosure and specific consent would be an inappropriate use of the technology.

Single player representation. The model reflects one player's behavioral patterns, which may encode individual habits, skill level, or cognitive style. Using predictions from this model to make decisions affecting other players (such as matchmaking, difficulty adjustment, or behavioral flagging) without validation on those players would be a misapplication.

Recommended use constraints. Appropriate uses of this model include research into player behavior modeling, prototyping adaptive game director systems with disclosed behavioral tracking, and academic study of action prediction from gameplay. Any production use should involve informed consent, multi subject validation, and transparent disclosure of what behavioral data is collected and how it is used.

Mitigation steps taken. The dataset (Atari-HEAD) was collected under informed consent from a research participant as part of a published study (Zhang et al., 2020). No personally identifiable information is included in the frame data. The model is evaluated on a held out test set to assess generalization. Its limitations are documented explicitly to prevent overconfident deployment.

Future Improvements

Hierarchical cascade classifier. The dominant remaining error pattern (NOOP vs. action) motivates a two stage architecture: Stage 1 trains a binary classifier on idle vs. active frames, and Stage 2 trains a four class directional classifier that only sees active frames. Each stage faces a simpler and better defined decision boundary, potentially eliminating the precision-recall tension that limits flat five class performance on NOOP.

Multi subject generalization. Training on gameplay from multiple players and evaluating cross player transfer would substantially increase the practical applicability of the approach. The Atari-HEAD dataset includes data from multiple subjects across other games; a multi subject extension using the same Montezuma's Revenge game would be a natural next step.

Longer temporal context. An LSTM or temporal attention module applied on top of the CNN feature extractor could capture behavioral sequences spanning more than four frames, enabling prediction of higher level intent such as navigating toward a key or retreating from an enemy.

Larger input resolution. Training at 160x210 (original ALE resolution) or with a ResNet feature extractor pretrained on natural images could improve spatial feature quality, particularly for distinguishing subtle enemy or object positions.

Future Integration

The player action predictor trained in this project could serve as a behavioral sensor module within a larger adaptive game director system. The stacked model produces a five class softmax output that estimates the probability of each action at each frame. A game director could consume this signal to detect behavioral patterns such as prolonged NOOP sequences indicating player confusion, repeated directional oscillation indicating navigation difficulty, or sustained UP and DOWN activity indicating active ladder traversal. These signals could trigger adaptive responses such as hint display, difficulty adjustment, or content modification.

The model's practical integration constraints should be acknowledged. It was trained on a single

player and would require fine-tuning or retraining on the target player population before deployment. Real time inference at 60fps is feasible on the RTX 3080 hardware used for training, but a lightweight deployment version would be needed for CPU or embedded environments. The five class action vocabulary is specific to Montezuma's Revenge and would need remapping for a different game title.

References

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